

Unsupervised Representation Learning for Images

Sunita Sarawagi

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Motivation

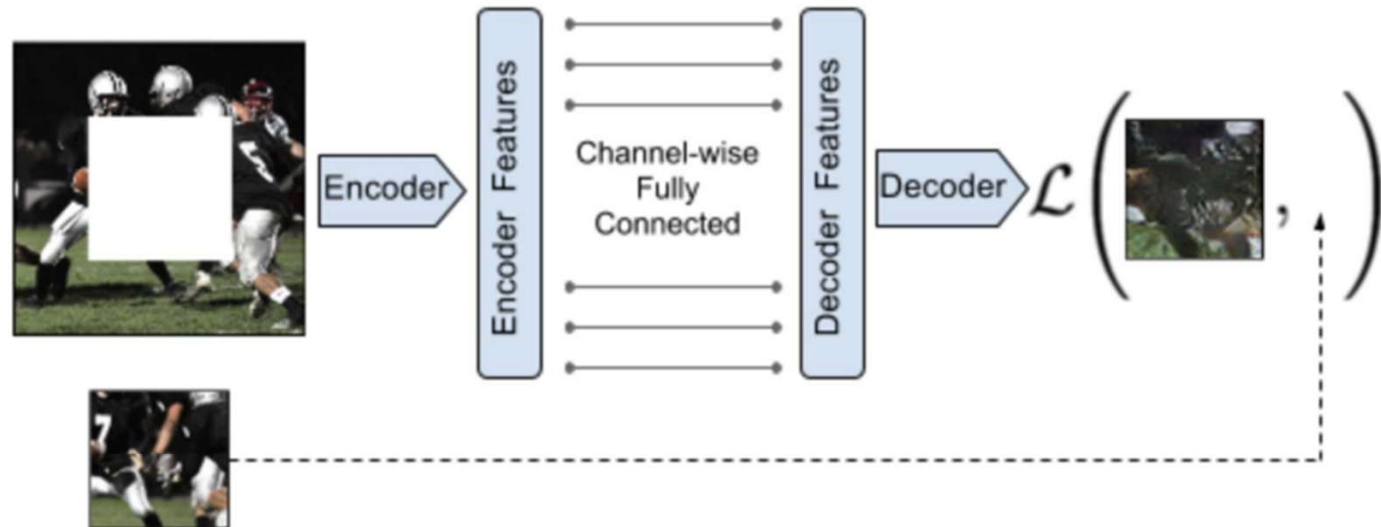
- ML models have more parameters than the amount of labelled data available for training them well
 - Labeled data: e.g.
 - Images along with labels of objects in them
 - Images with caption
- ResNET CNN with 50 layers has 23 million parameters
- Unlabelled data:
 - Large collection of images but without any labels or captions.
 - Can we harness these for pre-training a CNN for image classification or captioning?

Unsupervised → self-supervised

- Starting from unlabeled data e.g. collection of images, use a set of scripts to automatically create supervised tasks out of them.
- Example, for text data next-token prediction.
- Three types of self-supervised tasks for images:
 - Imputation
 - Mask part of image and generate that
 - Proxy or pretext tasks
 - Create image pairs and use Siamese networks to generate representations that can predict relationship between them
 - Contrastive learning
 - Like metric learning, but create similar pairs on your own.

Imputation

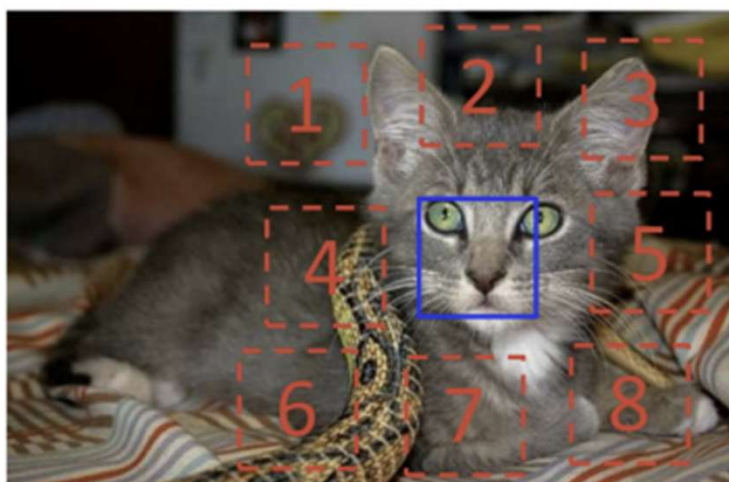
Learning to inpaint by reconstruction



Learning to reconstruct the missing pixels

Source: [Pathak et al., 2016](#)

Pretext task: predict relative patch locations



$$X = (\text{cat_face}, \text{cat_ear}); Y = 3$$

Example:



Question 1:

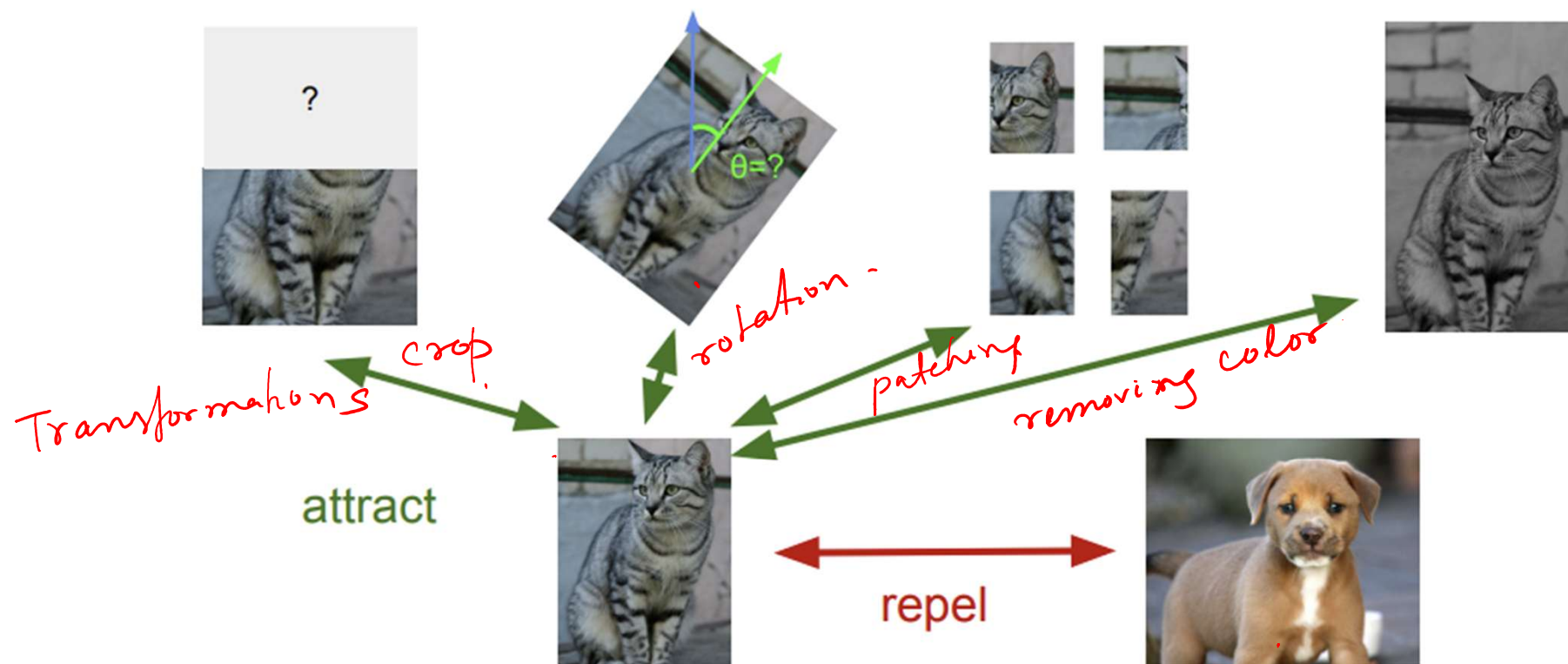


Question 2:

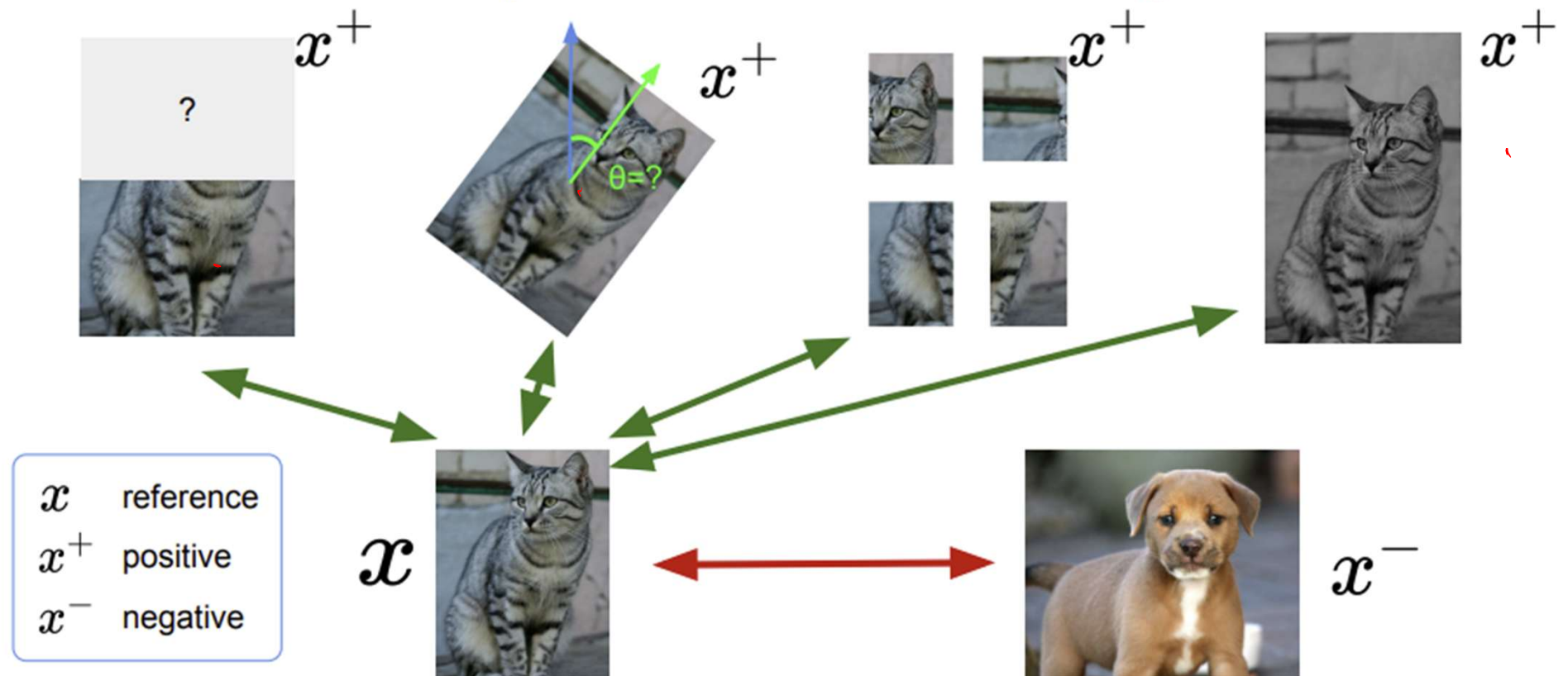


(Image source: [Doersch et al., 2015](#))

Contrastive Representation Learning



Contrastive Representation Learning



A formulation of contrastive learning

Loss function given 1 positive sample and $N - 1$ negative samples:

$$L = -\mathbb{E}_X \left[\log \frac{\exp(s(f(x), f(x^+)))}{\exp(s(f(x), f(x^+))) + \sum_{j=1}^{N-1} \exp(s(f(x), f(x_j^-)))} \right]$$

similarity function (pointing to s)
representation or embedding of input (pointing to f)



x



x^+



x



x_1^-



x_2^-



x_3^-

...

SimCLR: A Simple Framework for Contrastive Learning

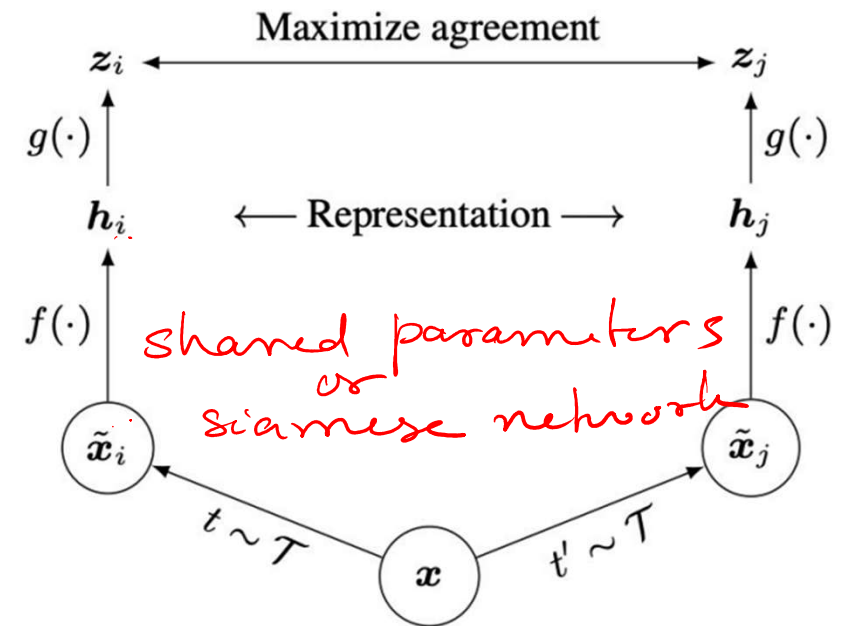
Cosine similarity as the score function:

$$s(u, v) = \frac{u^T v}{||u|| ||v||}$$

Use a projection network $g(\cdot)$ to project features to a space where contrastive learning is applied

Generate positive samples through data augmentation:

- random cropping, random color distortion, and random blur.



Source: [Chen et al., 2020](#)

SimCLR: generating positive samples from data augmentation



(a) Original



(b) Crop and resize



(c) Crop, resize (and flip)



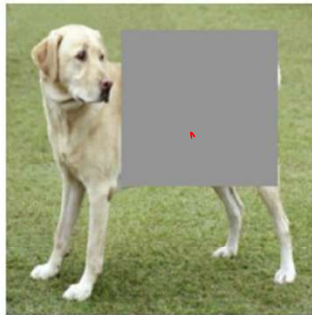
(d) Color distort. (drop)



(e) Color distort. (jitter)



(f) Rotate $\{90^\circ, 180^\circ, 270^\circ\}$



(g) Cutout



(h) Gaussian noise



(i) Gaussian blur



(j) Sobel filtering

Source: [Chen et al., 2020](#)

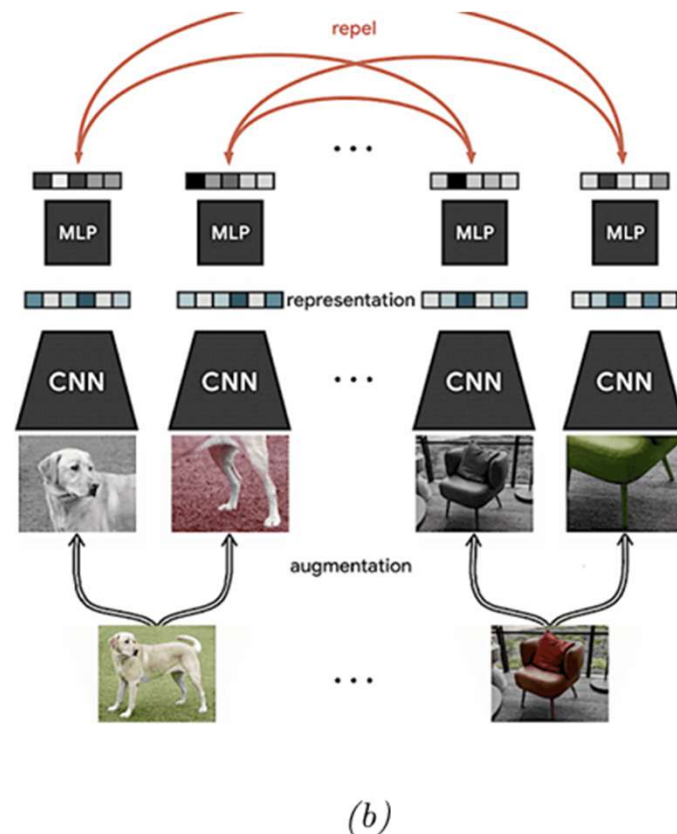
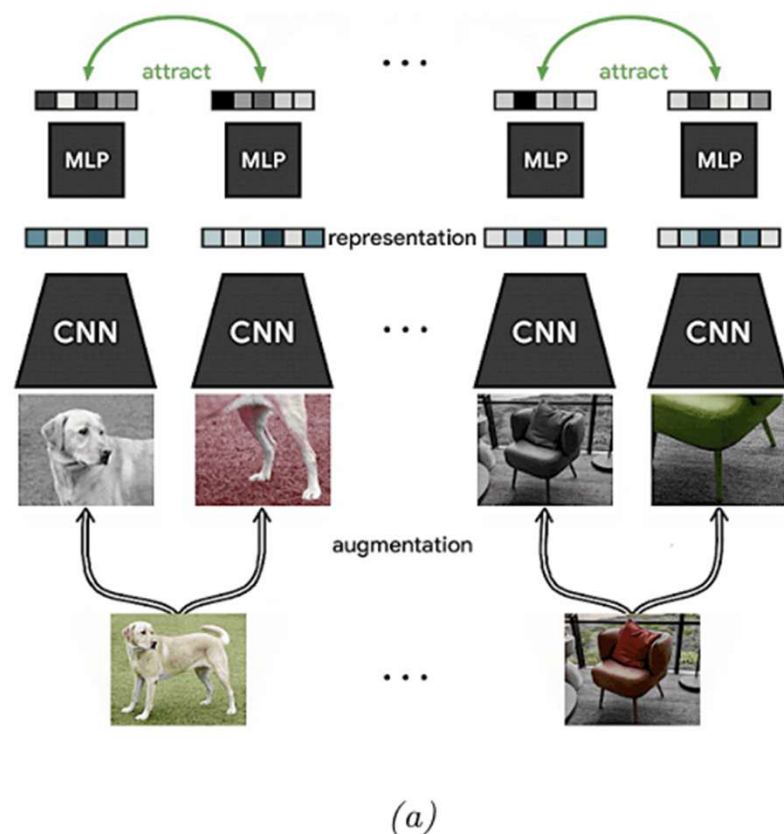


Figure 19.6: Visualization of SimCLR training. Each input image in the minibatch is randomly modified in two different ways (using cropping (followed by resize), flipping, and color distortion), and then fed into a Siamese network. The embeddings (final layer) for each pair derived from the same image is forced to be close, whereas the embeddings for all other pairs are forced to be far. From <https://ai.googleblog.com/2020/04/advancing-self-supervised-and-semi.html>. Used with kind permission of Ting Chen.

SimCLR

Algorithm 1 SimCLR's main learning algorithm.

input: batch size N , constant τ , structure of $f, g, \mathcal{T}$.
for sampled minibatch $\{\mathbf{x}_k\}_{k=1}^N$ **do**
 for all $k \in \{1, \dots, N\}$ **do**
 draw two augmentation functions $t \sim \mathcal{T}, t' \sim \mathcal{T}$
 # the first augmentation
 $\tilde{\mathbf{x}}_{2k-1} = t(\mathbf{x}_k)$
 $\mathbf{h}_{2k-1} = f(\tilde{\mathbf{x}}_{2k-1})$ # representation
 $\mathbf{z}_{2k-1} = g(\mathbf{h}_{2k-1})$ # projection
 # the second augmentation
 $\tilde{\mathbf{x}}_{2k} = t'(\mathbf{x}_k)$
 $\mathbf{h}_{2k} = f(\tilde{\mathbf{x}}_{2k})$ # representation
 $\mathbf{z}_{2k} = g(\mathbf{h}_{2k})$ # projection
 end for
 for all $i \in \{1, \dots, 2N\}$ and $j \in \{1, \dots, 2N\}$ **do**
 $s_{i,j} = \mathbf{z}_i^\top \mathbf{z}_j / (\|\mathbf{z}_i\| \|\mathbf{z}_j\|)$ # pairwise similarity
 end for
 define $\ell(i, j)$ as $\ell(i, j) = -\log \frac{\exp(s_{i,j}/\tau)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp(s_{i,k}/\tau)}$
 $\mathcal{L} = \frac{1}{2N} \sum_{k=1}^N [\ell(2k-1, 2k) + \ell(2k, 2k-1)]$
 update networks f and g to minimize $\mathcal{L}$
end for
return encoder network $f(\cdot)$, and throw away $g(\cdot)$

*We use a slightly different formulation in the assignment. You should follow the assignment instructions.

Generate a positive pair by sampling data augmentation functions

Iterate through and use each of the $2N$ sample as reference, compute average loss

InfoNCE loss:
Use all non-positive samples in the batch as $\mathbf{x}^-$

Source: [Chen et al., 2020](#)

Semi-supervised learning on SimCLR features

Method	Architecture	Label fraction	
		1%	10%
		Top 5	
Supervised baseline	ResNet-50	48.4	80.4
<i>Methods using other label-propagation:</i>			
Pseudo-label	ResNet-50	51.6	82.4
VAT+Entropy Min.	ResNet-50	47.0	83.4
UDA (w. RandAug)	ResNet-50	-	88.5
FixMatch (w. RandAug)	ResNet-50	-	89.1
S4L (Rot+VAT+En. M.)	ResNet-50 (4×)	-	91.2
<i>Methods using representation learning only:</i>			
InstDisc	ResNet-50	39.2	77.4
BigBiGAN	RevNet-50 (4×)	55.2	78.8
PIRL	ResNet-50	57.2	83.8
CPC v2	ResNet-161(*)	77.9	91.2
SimCLR (ours)	ResNet-50	75.5	87.8
SimCLR (ours)	ResNet-50 (2×)	83.0	91.2
SimCLR (ours)	ResNet-50 (4×)	85.8	92.6

Table 7. ImageNet accuracy of models trained with few labels.

Train feature encoder on **ImageNet** (entire training set) using SimCLR.

Finetune the encoder with 1% / 10% of labeled data on ImageNet.

Source: [Chen et al., 2020](#)

Implementation concerns

- Model learns meaningfully only with a sufficiently diverse set of negatives.
- To reduce cost of sampling negatives, one idea is to use a memory bank of past (negative) embeddings
 - Update embeddings using exponential moving averaging. This is known as momentum contrastive learning or MoCo [He+20]

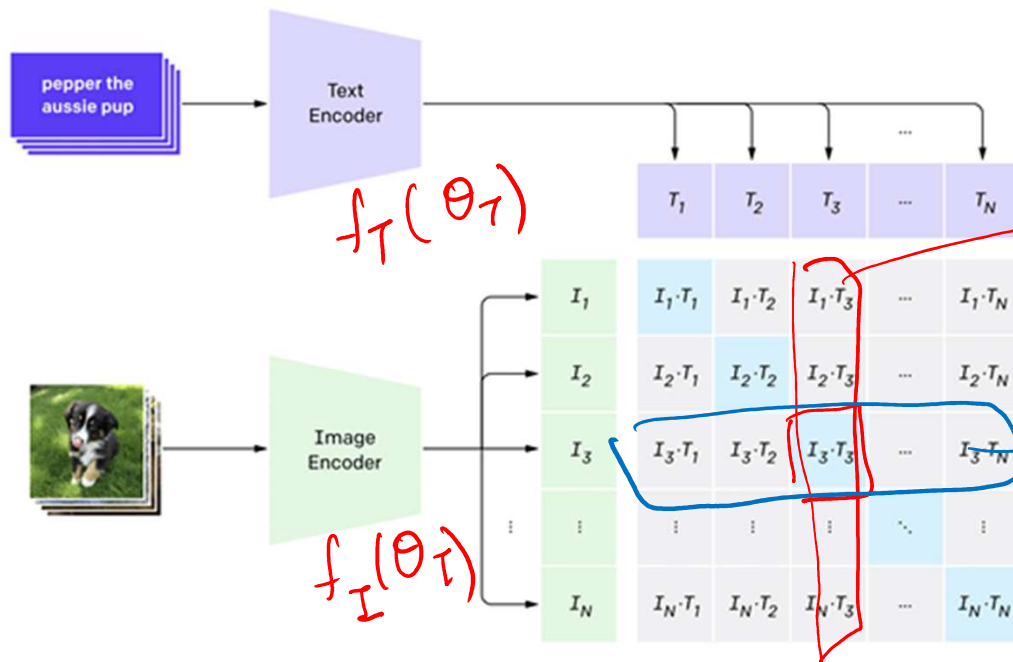
Self-supervised multi-modal pre-training

- Learning to jointly encode image and text
- Given loosely paired Text-Image pairs.

<https://icml.cc/media/icml-2021/Slides/9193.pdf>

CLIP: Contrastive Language-Image Pre-training

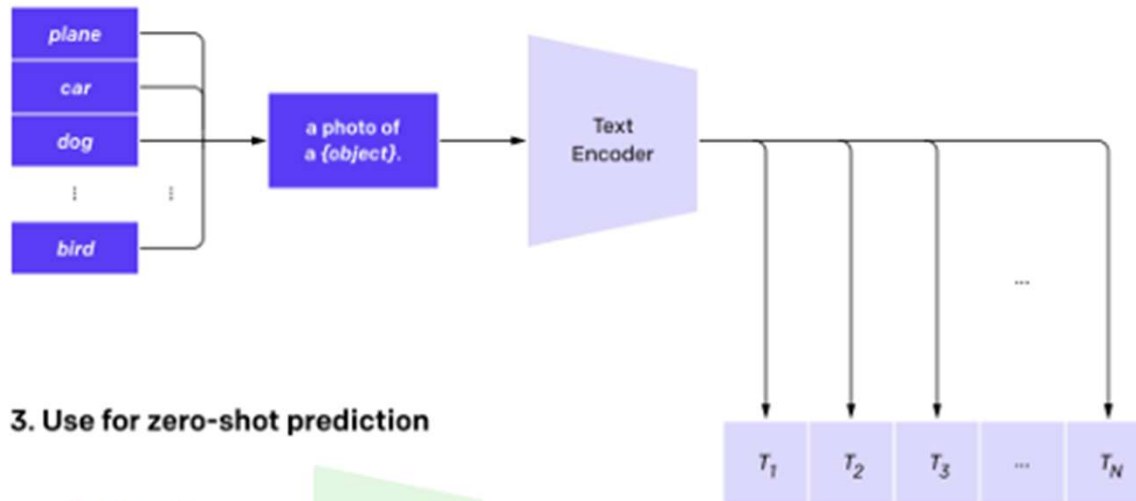
1. Contrastive pre-training



$$\min_{\theta_T, \theta_I} \sum_{i=1}^N \left(\log \frac{e^{T_i \cdot I_i}}{\sum_{j=1}^N e^{T_i \cdot I_j}} \right) + \log \frac{e^{T_i \cdot I_i}}{\sum_{j=1}^N e^{T_j \cdot I_i}}$$

Using CLIP for zero-shot classification

2. Create dataset classifier from label text



3. Use for zero-shot prediction

